

Jaynes-Cummings Hamiltonian

Open Quantum Systems

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Introduction

Let us consider a 2-level system in contact with a monochromatic external bath, such as a spin inside an electromagnetic cavity that admits a single frequency, or a superconducting qubit in a Josephson junction coupled to an LC circuit.

By quantizing the circuit, which is dissipationless and has a frequency depending on $\sqrt{LC}$, we obtain the Hamiltonian of a single-frequency harmonic oscillator. The total Hamiltonian will therefore be a 2×2 Hermitian matrix:

$$H = \frac{\omega}{2}\sigma_z + \omega_B b^\dagger b + g(\sigma^+ b + b^\dagger \sigma^-) \quad (1)$$

Where:

$$\sigma^\pm = \sigma_x \pm i\sigma_y \quad (2)$$

$$\sigma^+ \sigma^- = 2(1 + \sigma_z) \quad (3)$$

$$\sigma^- \sigma^+ = 2(1 - \sigma_z) \quad (4)$$

Since we are operating in a Hilbert space of the form $\mathcal{H} = \mathcal{H}_\omega \otimes \mathbb{C}^2$, the operators of the individual spaces commute with each other:

$$[b^\pm, \sigma^\pm] = 0 \quad (5)$$

Spectrum

We label the eigenstates of σ_z in the computational basis:

$$\{|0\rangle, |1\rangle\} \quad (6)$$

On this basis, the action of $\sigma^\pm = \sqrt{s(s+1) - m(m \pm 1)} = 1 \quad \forall m$.

For bosonic states, we use the Fock basis:

$$\hat{n} = b^\dagger b \quad (7)$$

$$\hat{n}|n\rangle = n|n\rangle \quad (8)$$

$$|n\rangle = \frac{1}{\sqrt{n!}} b^{+n} |0\rangle \quad (9)$$

We observe that H conserves the total number N , defined as $N = n + \{|0\rangle, |1\rangle\}$. Since they are compatible operators, we choose the basis of N :

$$\{|n\rangle \otimes \{|0\rangle, |1\rangle\}\} \quad (10)$$

Clearly, for a fixed $N + 1$ (we use $N + 1$ for convenience in notation), the only possible states are:

$$\begin{cases} |n, 1\rangle \\ |n + 1, 0\rangle \end{cases} \quad (11)$$

The idea is similar to $SU(2)$, where the Hamiltonian is infinite-dimensional but has a 2×2 block structure for a fixed $N + 1$. For $N \neq 0$, the individual blocks will be:

$$\begin{pmatrix} \langle N + 1, 0 | H | N + 1, 0 \rangle & \langle N + 1, 0 | H | N, 1 \rangle \\ \langle N, 1 | H | N + 1, 0 \rangle & \langle N, 1 | H | N, 1 \rangle \end{pmatrix} \quad (12)$$

The only 1×1 block left out is for $N = 0$:

$$(\langle 0, 0 | H | 0, 0 \rangle) \quad (13)$$

To find the spectrum, we require H to yield two energy levels on each block, except for the first block, which we will address later. Writing the action of H on the eigenstates yields a 2×2 matrix:

$$\begin{pmatrix} -\frac{\omega}{2} + \omega_B(N + 1) & g\sqrt{N + 1} \\ g\sqrt{N + 1} & \frac{\omega}{2} + \omega_B N \end{pmatrix} \quad (14)$$

Which has the following eigenvalue spectrum:

$$E_{\pm} = \frac{1}{2} \left(\omega_B(2N + 1) \pm \sqrt{4g^2(N + 1) + (\omega - \omega_B)^2} \right) \quad (15)$$

Now, let us interpret the initial 1×1 block as a boundary condition, since there is no physical reason why the energy spectrum should differ between the first block and all the others. The action of H on the ground state $|0, 0\rangle$ is:

$$H|0, 0\rangle = -\frac{\omega}{2}|0, 0\rangle \quad (16)$$

We impose that this be consistent with the solutions found earlier:

$$-\frac{\omega}{2} = \frac{1}{2} \left(\omega_B(2N + 1) \pm \sqrt{4g^2(N + 1) + (\omega - \omega_B)^2} \right) \quad (17)$$

After some calculations, we find that only the E_+ solution has physical meaning for the ground state block, because we obtain:

$$g^2 = \left(\frac{\omega_+ + \omega_-}{2} \right) \left(\frac{\omega_+ - \omega_-}{2} \right) > 0 \quad (18)$$

Where $\omega_+ = \omega + \omega_B$ and $\omega_- = \omega - \omega_B$.

The solution is obviously positive, meaning that g is real, just as we expect to ensure H is Hermitian. The E_- solution, on the other hand, would imply an imaginary g .

However, this solution should not be discarded, as it is consistent with the subsequent 2×2 block. Indeed, since the Hilbert space is infinite-dimensional, nothing prevents us from reorganizing the blocks by setting the bosonic excitations to 0 for the first 2×2 block, and to n excitations for the remaining 2×2 blocks. In that case, every block (including the ground state) would possess the $E_{\pm}$ solutions. Therefore, this is merely an arbitrary choice that, in our case, provides consistency conditions on g to make H Hermitian.

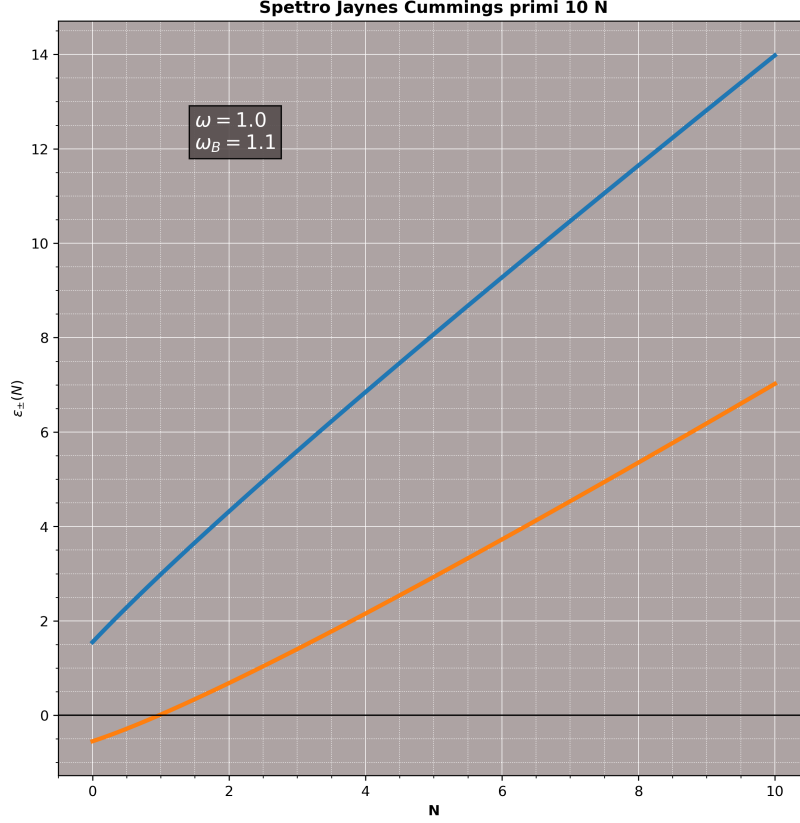


Figure 1: Spectrum as a function of N , evaluated at arbitrary but closely spaced values of ω and ω_B ($\omega = 1.0, \omega_B = 1.1$)

Lindblad

Let us now introduce Lindbladian dissipation, with its associated operators—one for bosonic dissipation and the other for spin dissipation, respectively:

$$\begin{cases} L_1 = \gamma b \rho b^\dagger - \frac{\gamma}{2} \{b^\dagger b, \rho\} \\ L_2 = \gamma_a \sigma^- \rho \sigma^+ - \frac{\gamma_a}{2} \{\sigma^+ \sigma^-, \rho\} \end{cases} \quad (19)$$

Where $\{\cdot\}$ denotes the anticommutator, and we have two characteristic rates, γ_a and γ . The associated Lindblad equation will be:

$$\dot{\rho} = -i[H, \rho] + L_1 + L_2 \quad (20)$$

Before diving into the calculations, let us think about what is happening physically: we have opened the cavity to allow dissipation, meaning the total number of bosons decreases, allowing photon exchange outside the cavity. Furthermore, we expect the $|1\rangle$ states, being spin excitations, to decay into $|0\rangle$.

Let us move to the basis of the total number operator and write the density operator for a single block (nothing prevents us from allowing a mixed-state density operator; in that case, we simply take the following blocks, weighted by their probabilities):

$$\rho = \sum \rho_{ij} |i\rangle \langle j| \quad (21)$$

Where we use the computational basis states as indices, noting that with a fixed total number, the bosonic excitations are not independent. Therefore, setting:

$$\dot{\rho}_{ij} = -i[H, \rho_{ij}] + \gamma \langle i | b \rho b^\dagger | j \rangle - \frac{\gamma}{2} \langle i | \{b^\dagger b, \rho\} | j \rangle + \gamma_a \langle i | \sigma^- \rho \sigma^+ | j \rangle - \frac{\gamma_a}{2} \langle i | \{\sigma^+ \sigma^-, \rho\} | j \rangle \quad (22)$$

Let us focus on the $(N + 1)$ block, which we will label with the superscript " N ". Sparing an atrocious amount of calculation, we arrive at the set of coupled differential equations:

$$\left\{ \begin{array}{l} \dot{\rho}_{00}^{N+1} = ig\sqrt{N+1}(\rho_{01}^{N+1} - \rho_{10}^{N+1}) + \gamma(N+2)\rho_{00}^{N+2} - \gamma(N+1)\rho_{00}^{N+1} + \gamma_a\rho_{11}^{N+2} \\ \dot{\rho}_{11}^{N+1} = -ig\sqrt{N+1}(\rho_{01}^{N+1} - \rho_{10}^{N+1}) - (\gamma N + \gamma_a)\rho_{11}^{N+1} + \gamma(N+1)\rho_{11}^{N+2} \\ \dot{\rho}_{01}^{N+1} = \sqrt{(N+1)(N+2)}\rho_{01}^{N+2} - \frac{\gamma}{2}(2N+1)\rho_{01}^{N+1} \\ \quad - \frac{\gamma_a}{2}\rho_{01}^{N+1} + ig\sqrt{(N+1)}(\rho_{00}^{N+1} - \rho_{11}^{N+1}) + i(\omega - \omega_B)\rho_{01}^{N+1} \\ \dot{\rho}_{10}^{N+1} = \sqrt{(N+1)(N+2)}\rho_{10}^{N+2} - \frac{\gamma}{2}(2N+1)\rho_{10}^{N+1} \\ \quad - \frac{\gamma_a}{2}\rho_{10}^{N+1} - ig\sqrt{(N+1)}(\rho_{00}^{N+1} - \rho_{11}^{N+1}) - i(\omega - \omega_B)\rho_{10}^{N+1} \end{array} \right. \quad (23)$$

We can enforce the condition that $\rho^+ = \rho$ is Hermitian, meaning the populations (regardless of the block) are real, while the coherences are complex conjugates. We can then introduce the angular variable for the $(N + 1)$ block:

$$ig\sqrt{N+1}(\rho_{01}^{N+1} - \rho_{10}^{N+1}) = 2g\sqrt{N+1}|\rho_{01}^{N+1}|\sin(\theta_{N+1}) \quad (24)$$

Where we exploited the fact that g is real. We also immediately notice that the differential equations for the coherences are conjugate to each other, as expected. The presence of imaginary terms in their time evolution indicates transient oscillations. We can thus rewrite the population equations in terms of the angular variables:

$$\left\{ \begin{array}{l} \dot{\rho}_{00}^{N+1} = 2g\sqrt{(N+1)}|\rho_{01}^{N+1}|\sin(\theta_{N+1}) + \gamma(N+2)\rho_{00}^{N+2} - \gamma(N+1)\rho_{00}^{N+1} + \gamma_a\rho_{11}^{N+2} \\ \dot{\rho}_{11}^{N+1} = -2g\sqrt{(N+1)}|\rho_{01}^{N+1}|\sin(\theta_{N+1}) - (\gamma N + \gamma_a)\rho_{11}^{N+1} + \gamma(N+1)\rho_{11}^{N+2} \\ \dot{\rho}_{01}^{N+1} = \sqrt{(N+1)(N+2)}\rho_{01}^{N+2} - \frac{\gamma}{2}(2N+1)\rho_{01}^{N+1} \\ \quad - \frac{\gamma_a}{2}\rho_{01}^{N+1} + ig\sqrt{(N+1)}(\rho_{00}^{N+1} - \rho_{11}^{N+1}) + i(\omega - \omega_B)\rho_{01}^{N+1} \\ \dot{\rho}_{10}^{N+1} = \sqrt{(N+1)(N+2)}\rho_{10}^{N+2} - \frac{\gamma}{2}(2N+1)\rho_{10}^{N+1} \\ \quad - \frac{\gamma_a}{2}\rho_{10}^{N+1} - ig\sqrt{(N+1)}(\rho_{00}^{N+1} - \rho_{11}^{N+1}) - i(\omega - \omega_B)\rho_{10}^{N+1} \end{array} \right. \quad (25)$$

This formulation could be useful because if the equations for the coherences admit oscillating complex solutions, we might also find a relation for the angle in the complex plane, allowing us to describe the dynamics in terms of this angle.

Now we just have to impose the dynamics on the Ground State:

$$\dot{\rho}_{00}^0 = \gamma\rho_{00}^1 + \gamma_a\rho_{11}^1 \quad (26)$$

Which is consistent with the other equations, provided we set $N = -1$ (which nothing prevents us from doing). We also note that it is strictly increasing, since the populations are positive-definite.

Before (attempting) to solve these equations, let us reason about them. First, notice that the Lindbladians (as expected) do not conserve the total number, meaning $[L_i, N] \neq 0 \quad \forall i$. This is because terms belonging to other blocks appear in the differential equations. Furthermore, we want the evolution to be completely positive and trace-preserving, meaning populations must be positive-definite, and the total trace (of the entire system, not the single block) must be $\text{Tr}(\rho) = 1$, which also implies $\text{Tr}(\dot{\rho}) = 0$. We have thus found a constraint.

We then observe that these equations hold true for any of the infinite blocks, since there is no reason for them to evolve with different physics. Focusing on a single block, we observe an exponentially decaying contribution due to itself, balanced by growing contributions proportional to the populations of subsequent blocks, which are positive-definite.

Therefore, we can imagine that populations in successive blocks tend to decrease, but this only holds if we introduce a cutoff, i.e., a maximum number of excitations; otherwise, while the fixed points wouldn't change, there would be contributions from infinite blocks. Additionally, the spin-up population $|\uparrow\rangle$ receives no contributions from the spin-down population $|\downarrow\rangle$.

In terms of diagrammatics, we can picture transitions occurring from spin-up excited states to spin-down states, but not vice versa. All of this suggests that the ground state $|0,0\rangle$ is a fixed point—meaning everything converges there, leading to the time limit $\rho_{00}^0(\infty) = 1$. This would be ideal, if not for the angular contribution from $\sin(\theta_{N+1})$ arising from the evolution of the coherences. We need to understand how this angle evolves, and especially whether it converges to 0 after a certain time interval or continues to oscillate.

Let us think about how to introduce the cutoff. A first approach would be to set a physical cutoff beyond which we cannot go. Surely, regardless of the nature of the bath, we are constrained by the Planck length, so I could define a maximum number of excitations (maximum frequency). But that doesn't get us very far. Another idea would be to simply introduce temperature and constrain the fermions to Fermi-Dirac statistics:

$$\frac{1}{1 + \exp \frac{(E-\mu)}{k_B T}} \quad (27)$$

And impose that at low temperatures T , there exists a maximum energy scale beyond which the distribution zeroes out the number of fermions. The problem is that this constraint is valid for a solid-state system, but not necessarily for an electromagnetic cavity.

Another constraint could arise in the case of a superconducting qubit coupled to an LC circuit, because there are energy scales that constrain Macroscopic Quantum Tunneling (MQT) processes. Again, the problem is that this would not apply to an electromagnetic cavity. We need a universal cutoff, intrinsic to the system. Our system has 4 frequencies (5, but g is bound by the boundary condition): ω , ω_B , and the rates γ , γ_a . This doesn't help us.

Honestly, the most promising idea involves temperature, imposing a constraint like $\hbar\omega N = \mu$. The problem is that using typical scales ($\omega \simeq \text{GHz}$, $\mu \simeq \text{eV}$) would still yield a massive constraint, $N \simeq N_A$.

Let us now assume that a cutoff exists, and let us call it N . Suppose we can initialize the state right in the $(N+1)$ block as a pure state. At the initial instant, by imposing the trace constraint and writing the equation separating the populations of one block from the next, we get:

$$\gamma(N+1)\rho_{00}^{N+1} + (\gamma N + \gamma_a)\rho_{11}^{N+1} = \gamma(N+2)\rho_{00}^{N+2} + \gamma_a\rho_{11}^{N+2} + \gamma(N+1)\rho_{11}^{N+2} \quad (28)$$

But at the initial instant, since only the $(N+1)$ block is populated, everything else is 0. Then, imposing $\rho_{11}^{N+1} = 1 - \rho_{00}^{N+1}$ (valid, again, only at the initial instant), we obtain:

$$\rho_{00}^{N+1} = \frac{\gamma N}{\gamma_a - \gamma} \quad (29)$$

Now, enforcing the condition that populations must be positive-definite implies that $\gamma_a > \gamma$. Furthermore, we require the populations to be ≤ 1 to find a constraint on N , which defines our cutoff:

$$N \leq \frac{\gamma_a - \gamma}{\gamma} \quad (30)$$

Python Simulation for Stationary Points

Now that we have found a cutoff, we notice an important feature: the system of ODEs is homogeneous, so we can treat it in matrix form and look for trivial eigenvectors. These should yield the stationary solution. The idea is to write a matrix separating the real and imaginary parts, except for the first block which we know to be real.

Thus, in the first step, the dimension doubles from $(N+1) \times (N+1)$ to $(2N+1) \times (2N+1)$. The problem now is that we cannot distinguish the different components of the density matrix within the

2×2 block; we can only distinguish real parts from imaginary parts. Therefore, the next step is to separate these components again, this time increasing the dimension to $(8N+1) \times (8N+1)$. Writing this large coefficient matrix so that it returns the matrix form (where the vectors have dimension $8N+1$):

$$A\rho = \dot{\rho} \quad (31)$$

By imposing a trivial solution ($\dot{\rho} = 0$), the problem is transformed into solving a system of linear equations. Having completed this step, we are now ready to solve the system, applying the following criteria to the solution: we round to 3 decimal places (completely arbitrary), treating everything else as noise; we verify that the trace equals 1; we verify that the populations are real and positive-definite; and we verify that the coherences within a single block are complex conjugates of each other.

I simulated the system with various parameters, and after filtering out various solutions, I find that the only stationary point is always the ground state. Interestingly, I only find it if I input the exact cutoff rounded up; if I round it down, it sometimes yields a ground state solution, but with a population of -1 . I don't know why.

With Driving Terms

We are given a set of driving terms to add to the Hamiltonian, namely:

$$\begin{cases} \omega_x \sigma_x \\ \Delta(b + b^+) \\ g\sigma_x(b + b^+) \end{cases} \quad (32)$$

For now, let's not apply them all together, and limit ourselves to studying the case where the qubit can flip, meaning only with $\omega_x \sigma_x$:

$$H = \frac{\omega}{2} \sigma_z + \omega_B b^+ b + g(\sigma^+ b + b^+ \sigma^-) + \omega_x \sigma_x \quad (33)$$

The problem now is that H no longer conserves the excitation number, since σ_x can flip a spin state. However, we can write $\sigma_x = \frac{\sigma^+ + \sigma^-}{2}$, which gives:

$$H = \frac{\omega}{2} \sigma_z + \omega_B b^+ b + g(\sigma^+ b + b^+ \sigma^-) + \frac{\omega_x}{2} (\sigma^+ + \sigma^-) \quad (34)$$

The idea is that we decouple the dynamics due to the baseline Jaynes-Cummings interaction, and then treat this driving term as something that merely makes the spins "jump" from one block to an adjacent one. We might already expect that, depending on the frequency ω_x , spin-up states will not necessarily converge into spin-down states.

From an immediate diagrammatic analysis, we can see jumps of the type:

$$\rho_{00}^{N+1} \rightarrow \rho_{01}^N \quad \& \quad \rho_{00}^{N+1} \rightarrow \rho_{10}^{N+2} \quad (35)$$

Because while it is true that on a given ket only one of the two $\sigma_{\pm}$ does not annihilate, its Hermitian conjugate can act simultaneously on the bra, meaning it transitions in two possible ways. This happens for every element of ρ , but fortunately, coherences can only jump into populations. This kick therefore kills populations but mixes coherences. I think we should simulate it.

Practically the same thing happens with the term $\Delta(b + b^+)$, but with the substantial difference that, except at the boundaries, these objects never annihilate, leading to transitions of every kind. I have a hunch that dynamic loops might emerge instead of true fixed points. Moving to field quadratures, these act as a position operator, but it would be better to ask theorists what can be said in QFT. This needs to be simulated too.

The last term, however, might be more interesting because it can be expanded as a product:

$$g\sigma_x(b + b^+) = \frac{g}{2} [(\sigma^+ b + b^+ \sigma^-) + (\sigma^- b + b^+ \sigma^+)] \quad (36)$$

Where I exploited the fact that they commute. Here, the first term is identical to the baseline Jaynes-Cummings interaction term but with a different rate, while the second term resembles the BCS term in mean-field theory, which could be expanded in a sort of Nambu notation.

The problem here is that no hole-particle symmetry appears like in BCS, because we are dealing with bosonic excitations on one side and fermionic ones on the other. However, we could quantize (and bosonize) the spin oscillations around an equilibrium position—an eigenstate of σ_z —using a Holstein-Primakoff method. This would allow us to rewrite it in Nambu notation to study the excitations as if we were in a superconducting transport system with Andreev processes.

After a few calculations, I couldn't make it work, so... let's just simulate it.